

Class 14: RNASeq mini-project

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Background

Data Import

Read counts and metadata CSV files

```
library(DESeq2)
```

```
Warning: package 'DESeq2' was built under R version 4.5.2
```

```
Warning: package 'S4Vectors' was built under R version 4.5.3
```

```
Warning: package 'BiocGenerics' was built under R version 4.5.2
```

Warning: package 'IRanges' was built under R version 4.5.2

Warning: package 'GenomicRanges' was built under R version 4.5.2

Warning: package 'Seqinfo' was built under R version 4.5.2

Warning: package 'SummarizedExperiment' was built under R version 4.5.2

Warning: package 'MatrixGenerics' was built under R version 4.5.2

Warning: package 'Biobase' was built under R version 4.5.3

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
              length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
              SRR493371
ENSG00000186092         0
ENSG00000279928         0
```

ENSG00000279457	46
ENSG00000278566	0
ENSG00000273547	0
ENSG00000187634	258

Sanity Check

Q. Complete the code below to remove the troublesome first column from countData

```
countData <- as.matrix(countData[, -1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
countData = countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Setup DESeq object

```
metadata
```

standardGeneric for "metadata" defined from package "S4Vectors"

```
function (x, ...)
standardGeneric("metadata")
<bytecode: 0x111869340>
<environment: 0x11185ff58>
Methods may be defined for arguments: x
Use showMethods(metadata) for currently available ones.
```

```
library(DESeq2)
```

```
dds = DESeqDataSetFromMatrix(countData = countData,
                             colData = colData,
                             design = ~ condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```
res = results(dds)
```

Run DESeq analysis

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Extract the results

Big table with log2 fold changes and p-values

```
res <- results(dds, contrast = c("condition", "hoxa1_kd", "control_sirna"))
head(res)
```

log2 fold change (MLE): condition hoxa1_kd vs control_sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76553e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
ENSG00000187642	4.03379e-01				

Data visualization

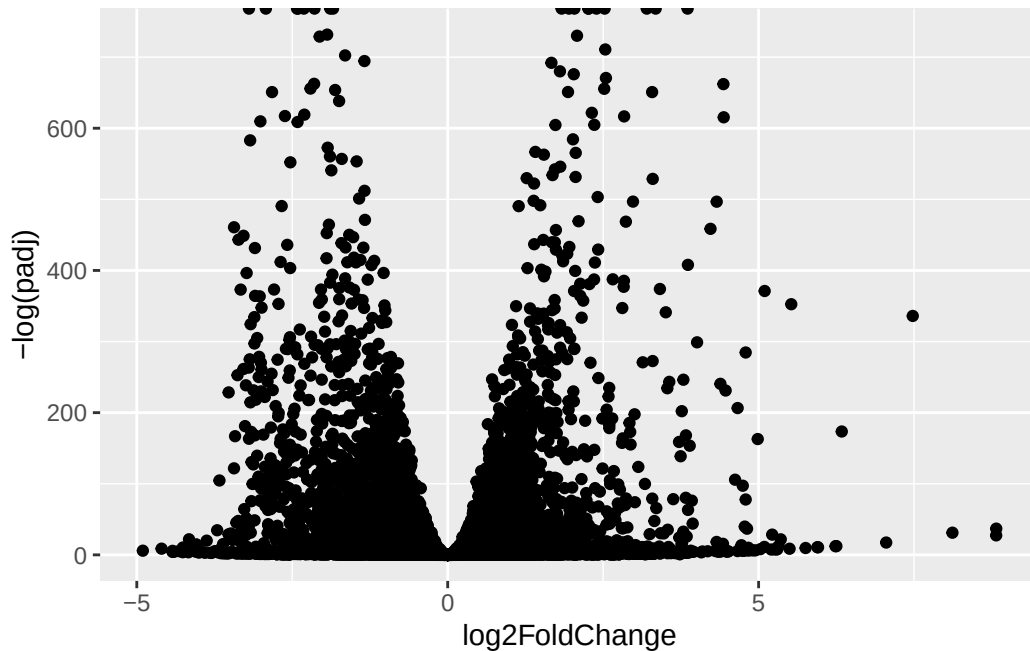
Volcano plot

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
ggplot(res) +  
  aes(log2FoldChange, -log(padj)) +  
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Make a color vector for all genes
mycols <- rep("orange", nrow(res))

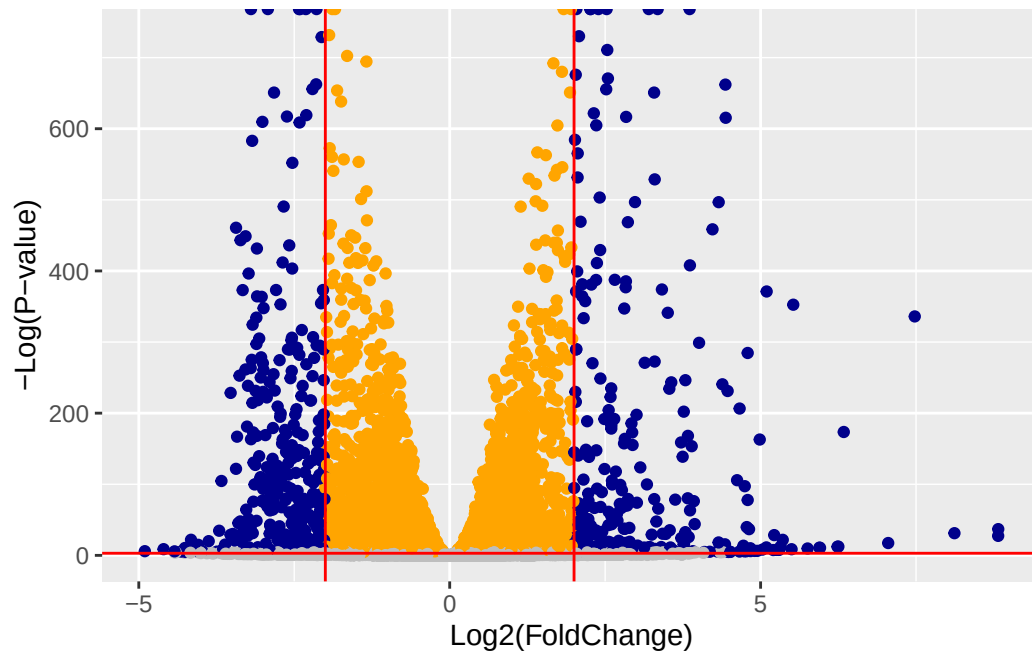
# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "darkblue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col=mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2), col = "red") +
  geom_hline(yintercept = -log(0.05), col = "red")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range

```
(`geom_point()`).
```



Add annotation data

Add gene symbol and entrez ids

Q. Use the `mapIDs()` function multiple times to add `SYMBOL`, `ENTREZID` and `GENENAME` annotation to our results by completing the code below.

```
library("AnnotationDbi")
```

Warning: package 'AnnotationDbi' was built under R version 4.5.2

```
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```



```

[1] "ACCNUM"      "ALIAS"      "ENSEMBL"    "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"     "EVIDENCE"   "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"         "GOALL"      "IPI"           "MAP"
[16] "OMIM"        "ONTOLOGY"   "ONTOLOGYALL" "PATH"          "PFAM"
[21] "PMID"        "PROSITE"    "REFSEQ"     "SYMBOL"        "UCSCKG"
[26] "UNIPROT"

```

```

res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")

```

'select()' returned 1:many mapping between keys and columns

```

res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")

```

'select()' returned 1:many mapping between keys and columns

```

res$name = mapIds(org.Hs.eg.db,
                  keys=row.names(res),
                  keytype="ENSEMBL",
                  column="GENENAME",
                  multiVals="first")

```

'select()' returned 1:many mapping between keys and columns

```

head(res, 10)

```

log2 fold change (MLE): condition hoxa1_kd vs control_sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03

ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.86555e-01	NA	NA		NA
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGRN	375790		agrin
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")
```

Pathway Analysis

KEGG, GO and REACTOME

KEGG

```
library(pathview)
```

Warning: package 'pathview' was built under R version 4.5.2

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

Warning: package 'gage' was built under R version 4.5.2

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```

[1] "100"      "10201"    "10606"    "10621"    "10622"    "10623"    "107"      "10714"
[9] "108"      "10846"    "109"      "111"      "11128"    "11164"    "112"      "113"
[17] "114"      "115"      "122481"   "122622"   "124583"   "132"      "158"      "159"
[25] "1633"     "171568"   "1716"     "196883"   "203"      "204"      "205"      "221823"
[33] "2272"     "22978"    "23649"    "246721"   "25885"    "2618"     "26289"    "270"
[41] "271"      "27115"    "272"      "2766"     "2977"     "2982"     "2983"     "2984"
[49] "2986"     "2987"     "29922"    "3000"     "30833"    "30834"    "318"      "3251"
[57] "353"      "3614"     "3615"     "3704"     "377841"   "471"      "4830"     "4831"
[65] "4832"     "4833"     "4860"     "4881"     "4882"     "4907"     "50484"    "50940"
[73] "51082"    "51251"    "51292"    "5136"     "5137"     "5138"     "5139"     "5140"
[81] "5141"     "5142"     "5143"     "5144"     "5145"     "5146"     "5147"     "5148"
[89] "5149"     "5150"     "5151"     "5152"     "5153"     "5158"     "5167"     "5169"
[97] "51728"    "5198"     "5236"     "5313"     "5315"     "53343"    "54107"    "5422"
[105] "5424"     "5425"     "5426"     "5427"     "5430"     "5431"     "5432"     "5433"
[113] "5434"     "5435"     "5436"     "5437"     "5438"     "5439"     "5440"     "5441"
[121] "5471"     "548644"   "55276"    "5557"     "5558"     "55703"    "55811"    "55821"
[129] "5631"     "5634"     "56655"    "56953"    "56985"    "57804"    "58497"    "6240"
[137] "6241"     "64425"    "646625"   "654364"   "661"      "7498"     "8382"     "84172"
[145] "84265"    "84284"    "84618"    "8622"     "8654"     "87178"    "8833"     "9060"
[153] "9061"     "93034"    "953"      "9533"     "954"      "955"      "956"      "957"
[161] "9583"     "9615"

```

```

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)

```

```

      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288

```

Let's run the gage analysis

```

# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)

```

```

attributes(keggres)

```

```

$names
[1] "greater" "less"    "stats"

```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04110.pathview.png

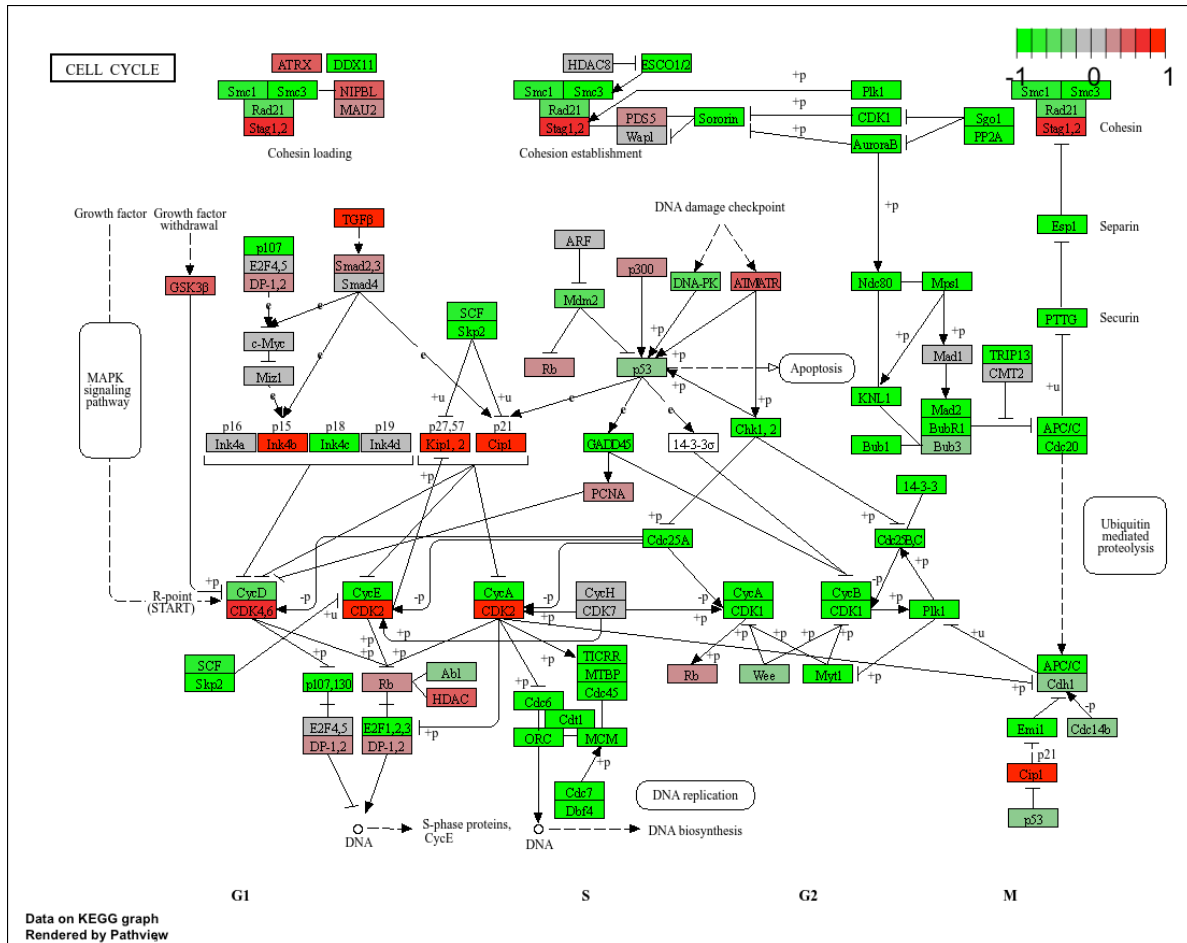


Figure 1: PNG

```
# change pathway, generate PDF graph
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
[,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04110.pathview.pdf

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggresdown <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggresdown, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

Finally, lets pass these IDs in `keggresids` to the `pathview()` function to draw plots for all the top 5 pathways.

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04330.pathview.png

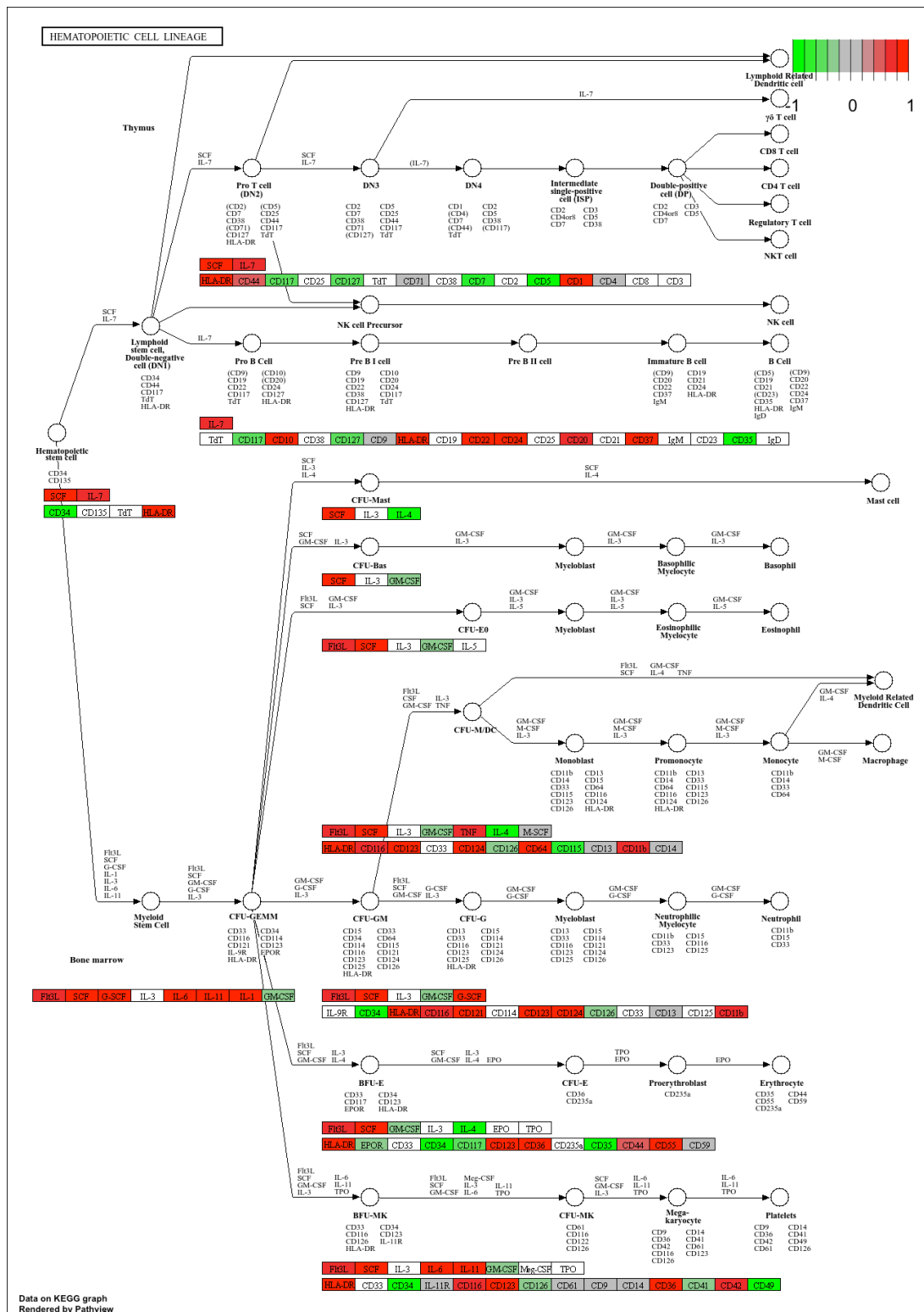


Figure 2: hsa04640

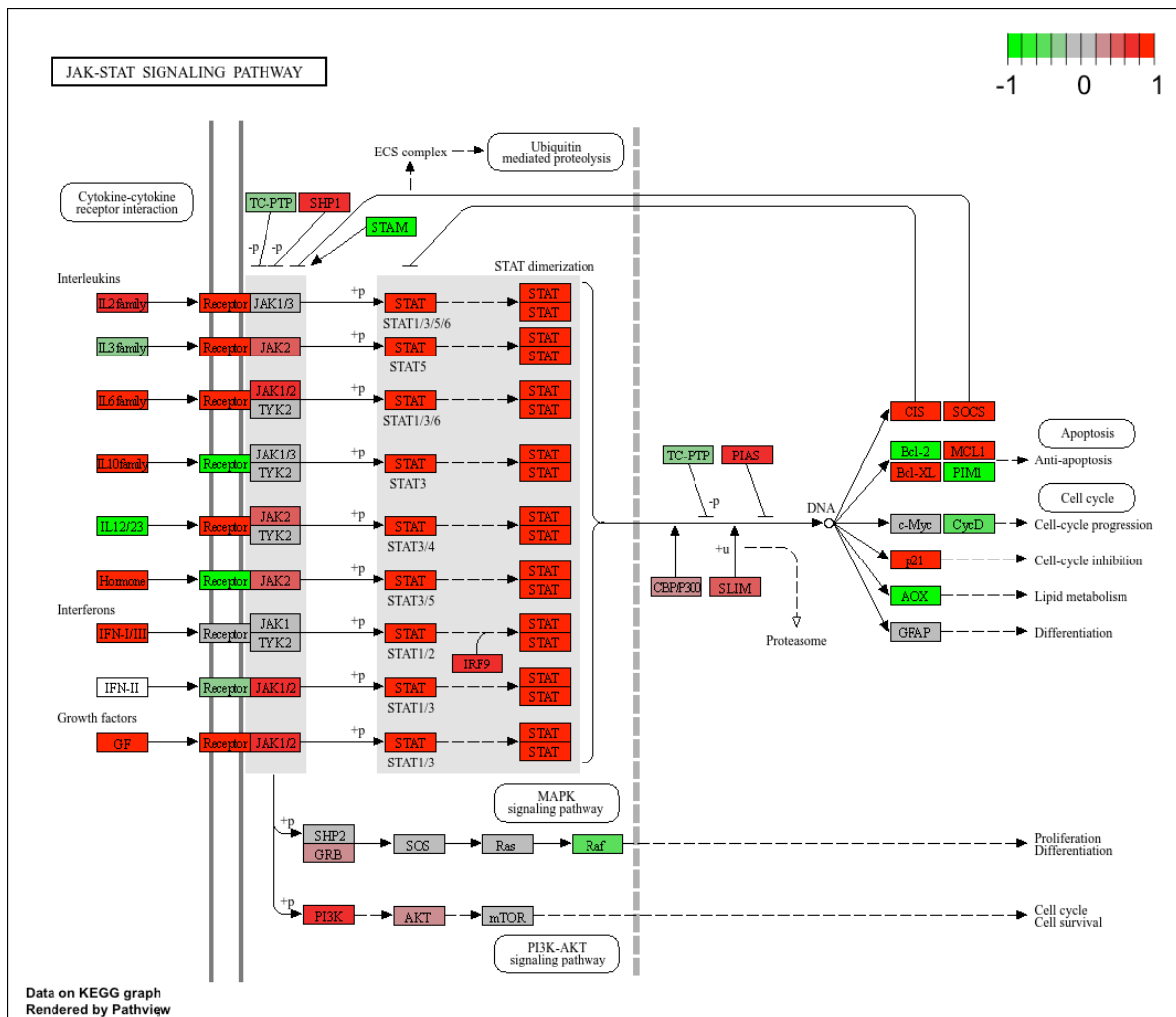


Figure 3: hsa04630

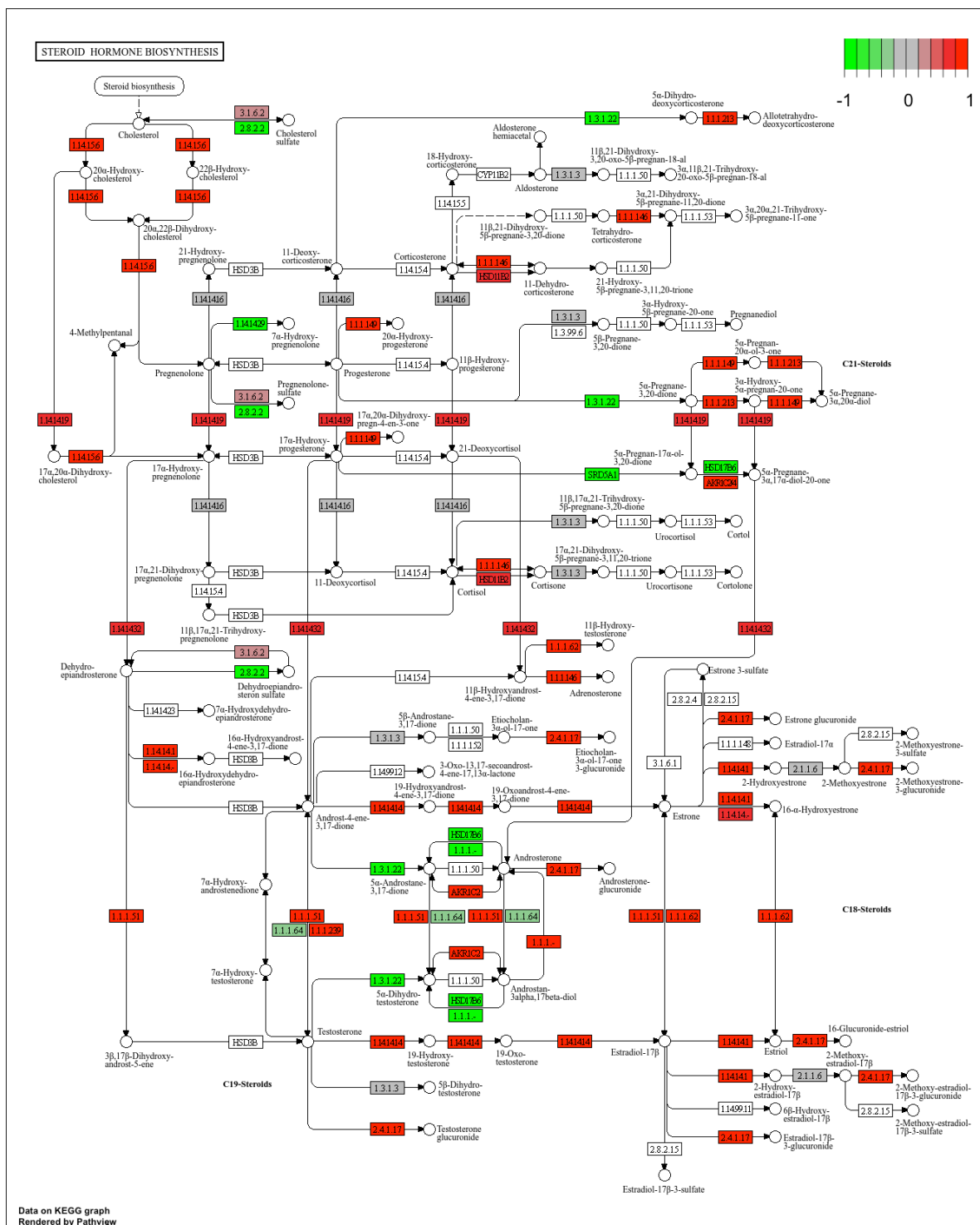


Figure 4: hsa00140

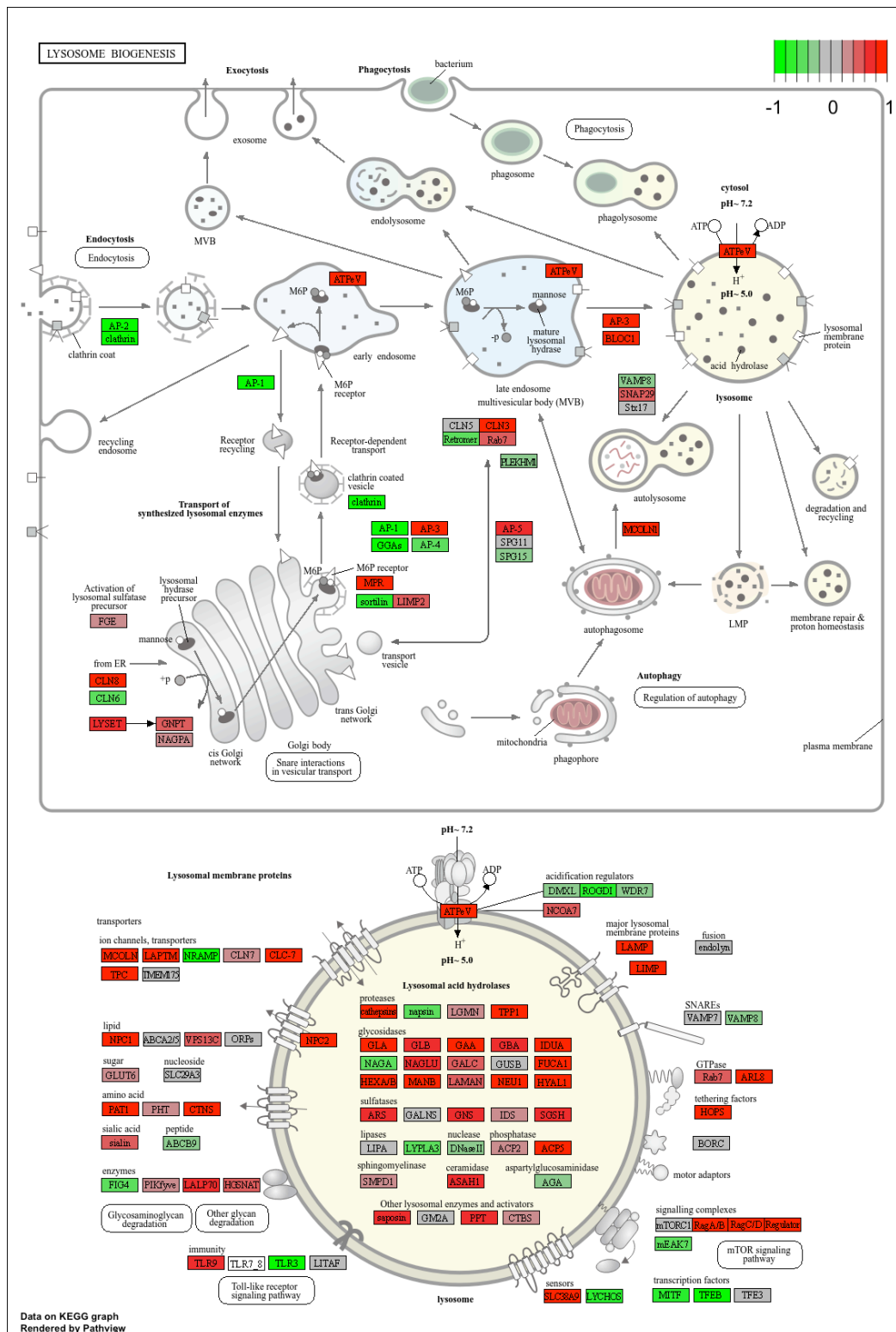


Figure 5: hsa04142

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
# Top 5 down-regulated KEGG pathways
keggrespathways_down <- rownames(keggres$less)[1:5]

keggresids_down <- substr(keggrespathways_down,
                          start = 1,
                          stop = 8)

keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

Draw pathview plots for the top 5 down-regulated pathways

```
pathview(gene.data = foldchanges,
         pathway.id = keggresids_down,
         species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/sofiajaravata/Desktop/bimm143/class14

Info: Writing image file hsa03013.pathview.png

Info: Writing image file hsa04114.pathview.png

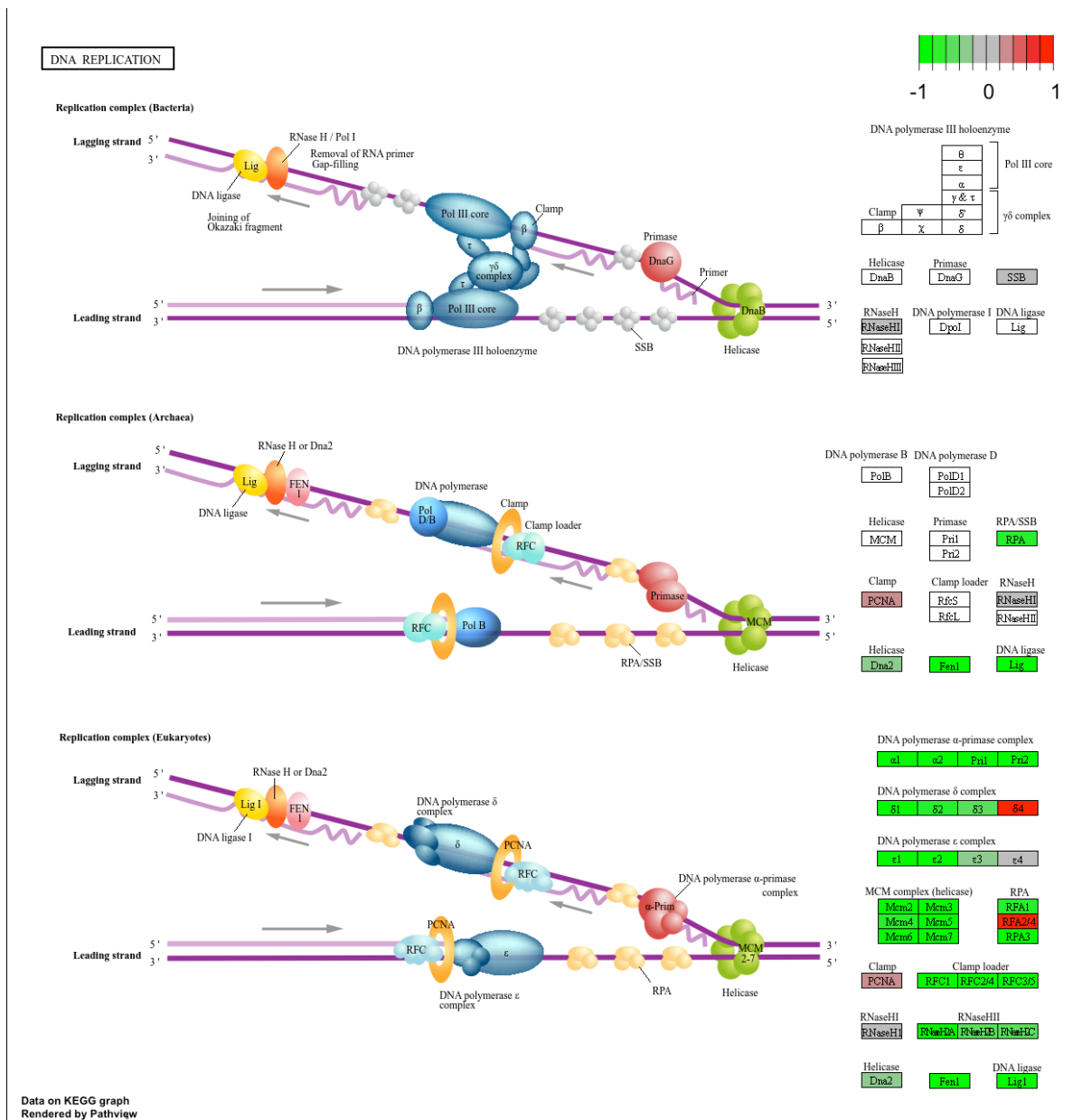


Figure 7: hsa03030

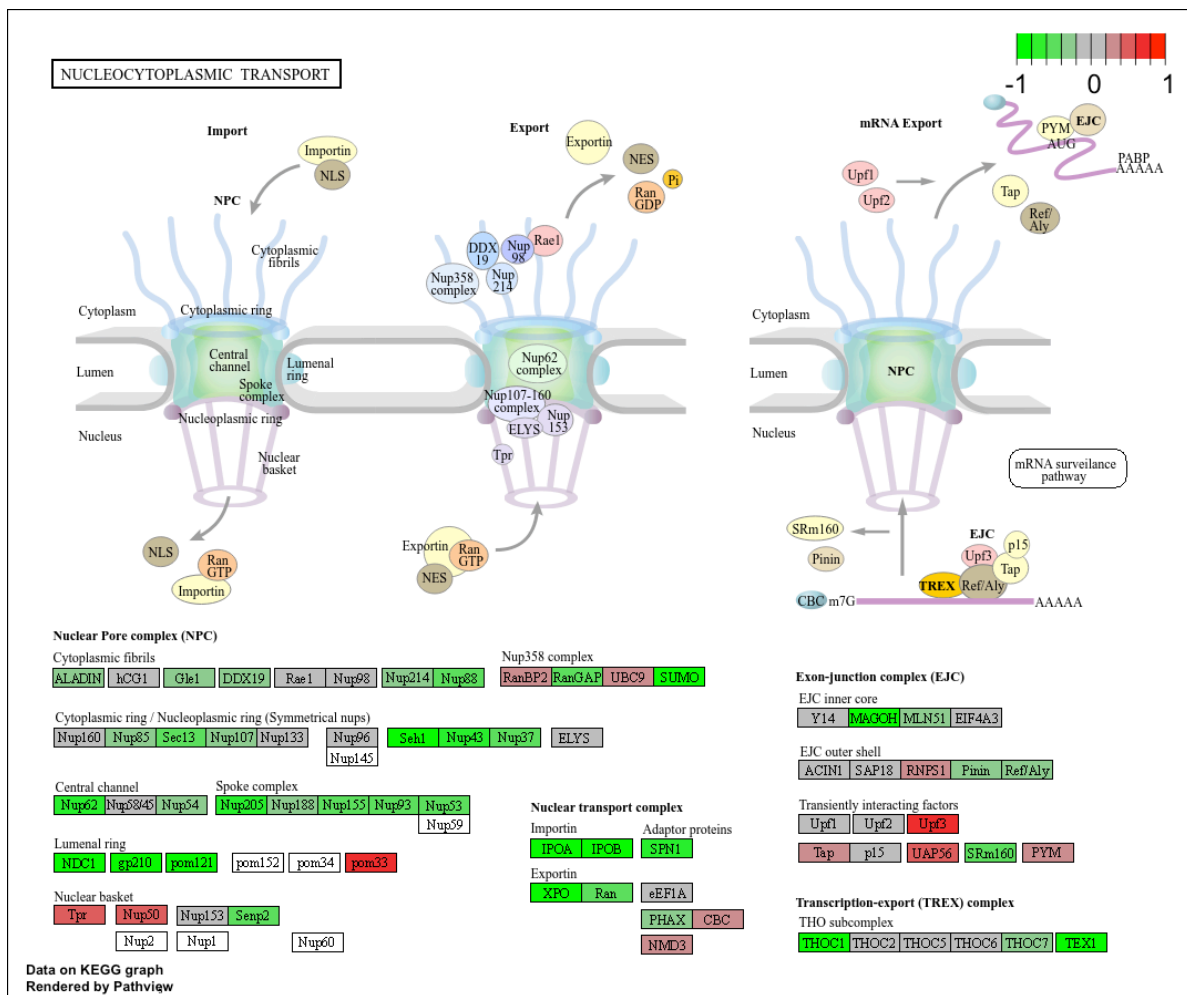


Figure 8: hsa03013

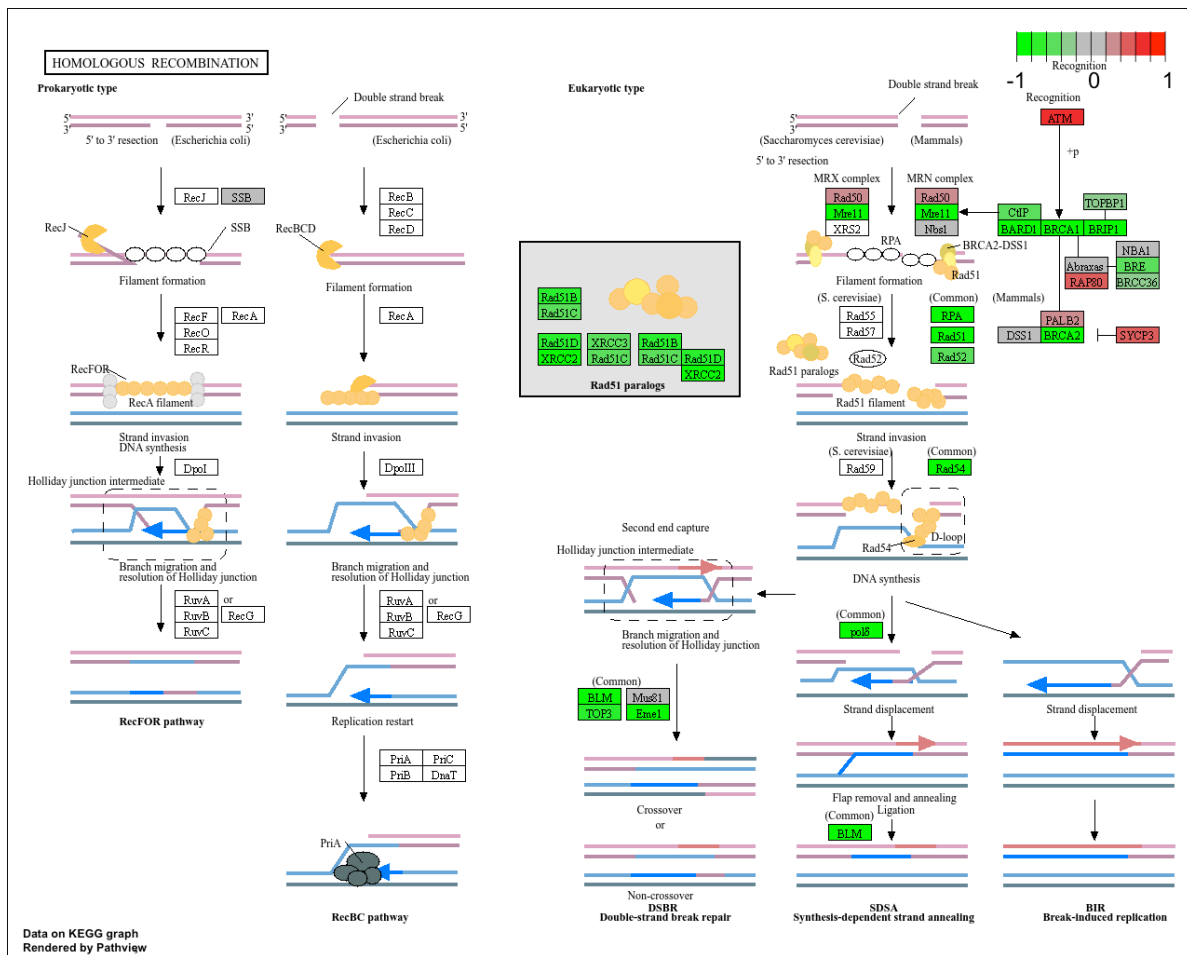


Figure 9: hsa03440

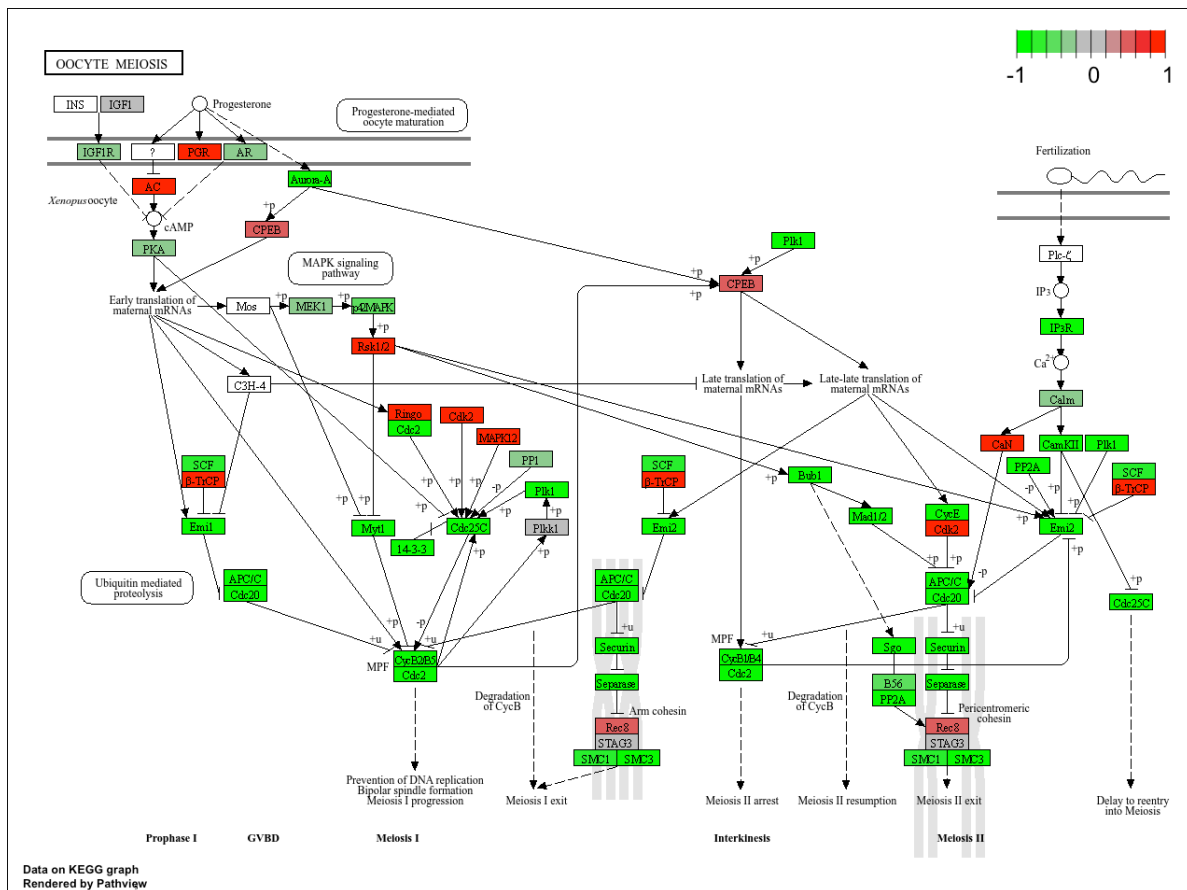


Figure 10: hsa04114

Gene Ontology (GO)

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04

	q.val	set.size	exp1
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	exp1
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432
G0:0060562 epithelial tube morphogenesis	3.261376	3.261376
G0:0035295 tube development	3.253665	3.253665

REACTOME

There is an R package for this analysis and a new-ish website <https://reactome.org/>

To use the website you can paste.

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=)
```

Question 8. What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway Cell Cycle ($p=2.64 \times 10^{-5}$), hsa04110, has the most significant “Entities p-value”. These most significant pathways match my previous KEGG results, identifying cell cycle as one of the most significant pathways alongside Cell Cycle, Mitotic ($p=2.78 \times 10^{-5}$). Some factors that could cause differences are the databases that define pathways differently, the genes included in each pathway, as well as the statistical methods used for analysis.

Save our results

```
#Save significant genes  
write.csv(sig_genes, file = "significant_genes.csv")  
  
# Save fold change vector  
write.csv(foldchanges, file = "foldchanges.csv")
```